



Ordering numbers in standard form

Task A: Comparing numbers in standard form

1) Put the following in ascending order, smallest to largest.

- a) 3×10^{12} 1×10^{13} 4×10^{11} 2×10^{10} 9×10^9
 9×10^9 2×10^{10} 4×10^{11} 3×10^{12} 1×10^{13}
- b) 2.3×10^4 2.3×10^{-1} 2.3×10^{-4} 2.3×10^{-3} 2.3×10
 2.3×10^{-4} 2.3×10^{-3} 2.3×10^{-1} 2.3×10 2.3×10^4
- c) 4.99×10^6 4×10^6 9.8×10^{-7} 4.99×10^{-6} 4.9×10^{-6}
 9.8×10^{-7} 4.9×10^{-6} 4.99×10^{-6} 4×10^6 4.99×10^6
- d) 9.7×10^{-2} 1.001×10^9 9.89×10^{-2} 1.5×10^{-4} 1.01×10^9
 1.5×10^{-4} 9.7×10^{-2} 9.89×10^{-2} 1.001×10^9 1.01×10^9

2) Put the following in ascending order in the format they are given.

- a) 1×10^3 0.0089 0.1 8.2×10^{-4} 90 000
 1×10^3 8.9×10^{-3} 1×10^{-1} 8.2×10^{-4} 9×10^4
 8.2×10^{-4} 0.0089 0.1 1×10^3 90 000
- b) 21 000 0.000 087 0.0002 2.01×10^4 0.000 078
 2.1×10^4 8.7×10^{-5} 2×10^{-4} 2.01×10^4 7.8×10^{-5}
0.000 078 0.000 087 0.0002 2.01×10^4 21 000

3) Laura says



Laura

I have put the following in ascending order by ordering the exponents first and where the exponents are the same, I have ordered the number.

Explain why Laura has not correctly ordered the numbers.

$$0.6 \times 10^{-1}$$

$$6 \times 10^2$$

$$6.2 \times 10^2$$

$$0.618 \times 10^3$$

To order according to the exponents and then the A number, all the numbers must be correctly written in standard form. Some of these numbers are not in standard form.



Task B: Ordering numbers using standard form

1) Using inequalities, $<$, $>$ or $=$, compare the sizes of each number, making sure to show your working out.

$$\text{a) } 38 \times 0.01 > 4.8 \times 10^{-2} \qquad 3.8 \times 10^{-1} > 4.8 \times 10^{-2}$$

$$\text{b) } 78\,000 \times 0.001 < 99 \times 10^2 \qquad 7.8 \times 10 < 9.9 \times 10^3$$

$$\text{c) } 139 \times \frac{1}{10^2} > 4820 \times \frac{1}{10^4} \qquad 1.39 \times 10^0 > 4.82 \times 10^{-1}$$

$$\text{d) } 1600 \times \frac{1}{10^2} = 160\,000 \times \frac{1}{10^4} \qquad 1.6 \times 10^1 = 1.6 \times 10^1$$

2) By working out the number, write in standard form and order the original form given in the question.

$$\begin{array}{cccc} \text{a) } & 3.8 \times 0.001 & 7900 \times 10^{-4} & 7.8 \times \frac{1}{10^2} & 800 \times \frac{1}{10^{-2}} \\ & 3.8 \times 10^{-3} & 7.9 \times 10^{-1} & 7.8 \times 10^{-2} & 8 \times 10^0 \\ & 3.8 \times 0.001 & 7.8 \times \frac{1}{10^2} & 7900 \times 10^{-4} & 800 \times \frac{1}{10^{-2}} \end{array}$$

$$\begin{array}{cccc} \text{b) } & 920 \times 0.0001 & 92\,100 \times 10^{-3} & 9.8 \times \frac{1}{10} & 9100 \times \frac{1}{10^{-2}} \\ & 9.2 \times 10^{-2} & 9.21 \times 10 & 9.8 \times 10^{-1} & 9.1 \times 10 \\ & 920 \times 0.0001 & 9.8 \times \frac{1}{10} & 9100 \times \frac{1}{10^2} & 92\,100 \times 10^{-3} \end{array}$$

$$\begin{array}{cccc} \text{c) } & 480 \times 0.1 & 4.8 \times 10^{-2} & 4.81 \times \frac{1}{10^2} & 4820 \times \frac{1}{10^4} \\ & 4.8 \times 10 & 4.8 \times 10^{-2} & 4.81 \times 10^{-2} & 4.82 \times 10^{-1} \\ & 4.8 \times 10^{-2} & 4.81 \times \frac{1}{10^2} & 4820 \times \frac{1}{10^4} & 480 \times 0.1 \end{array}$$